

Name – Earl Pratik Sinha

Roll – 2024AD05449

Email – 2024ad05449@wilp.bits-pilani.ac.in

MLOps Assignment 2:

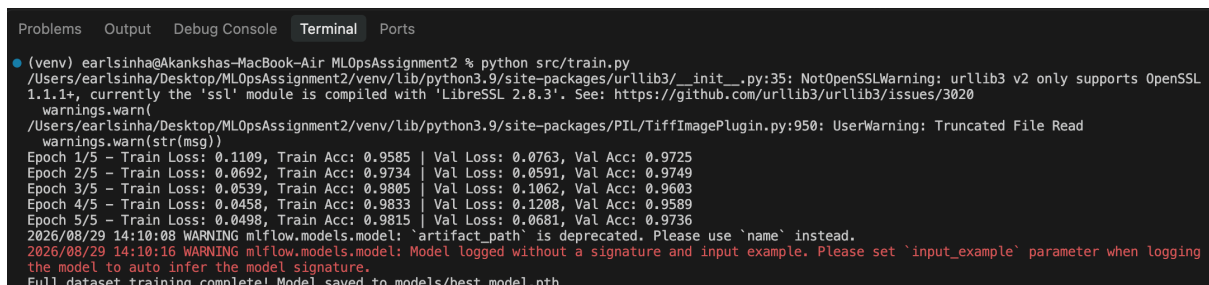
GitHub Repository: <https://github.com/earlpratiksinha/MLOpsAssignment2.git>

Video link: <https://drive.google.com/file/d/1DWfP0fI3VLb-j0xlTkk5GZbvYoVd9L-N/view?usp=sharing>

1. Model Training & Pipeline Verification

The model training was executed using PyTorch with transfer learning via MobileNetV2 over 5 training epochs.

- **Training Results:** Final Training Loss reached **0.0498** with a Training Accuracy of **98.15%**. Final Validation Loss landed at **0.0681** with a Validation Accuracy of **97.36%**.
- **Artifact Storage:** The trained model state dictionary was saved to `models/best_model.pth`.



```
(venv) earlsinha@Akankshas-MacBook-Air MLOpsAssignment2 % python src/train.py
/Users/earlsinha/Desktop/MLOpsAssignment2/venv/lib/python3.9/site-packages/urllib3/_init_.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL
1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
warnings.warn(
/Users/earlsinha/Desktop/MLOpsAssignment2/venv/lib/python3.9/site-packages/PIL/TiffImagePlugin.py:950: UserWarning: Truncated File Read
warnings.warn(str(msg))
Epoch 1/5 - Train Loss: 0.1109, Train Acc: 0.9585 | Val Loss: 0.0763, Val Acc: 0.9725
Epoch 2/5 - Train Loss: 0.0692, Train Acc: 0.9734 | Val Loss: 0.0591, Val Acc: 0.9749
Epoch 3/5 - Train Loss: 0.0539, Train Acc: 0.9805 | Val Loss: 0.1062, Val Acc: 0.9603
Epoch 4/5 - Train Loss: 0.0458, Train Acc: 0.9833 | Val Loss: 0.1208, Val Acc: 0.9589
Epoch 5/5 - Train Loss: 0.0498, Train Acc: 0.9815 | Val Loss: 0.0681, Val Acc: 0.9736
2026/08/29 14:10:08 WARNING mlflow.models.model: 'artifact_path' is deprecated. Please use 'name' instead.
2026/08/29 14:10:16 WARNING mlflow.models.model: Model logged without a signature and input example. Please set 'input_example' parameter when logging
the model to auto infer the model signature.
Full dataset training complete! Model saved to models/best_model.pth
```

2. Experiment Tracking with MLflow

All hyperparameter configurations, evaluation metrics, and model outputs were tracked using MLflow under the experiment run `awesome-gnat-576`.

Parameter / Metric	Tracked Value
Model Architecture	MobileNetV2
Dataset Size	17,498 images
Epochs / Batch Size	5 / 32
Learning Rate (lr)	0.001
Validation Accuracy	97.36%

mlflow
3.1.4
Experiments
Models
Prompts

Cat_Dog_Classification_Full >
awesome-gnat-576

Overview
Model metrics
System metrics
Traces
Artifacts

Metrics (4)

Q Search metrics

Metric	Value	
val_loss	0.06806579866930842	
train_acc	0.9815407475140016	0.06806579866930842
train_loss	0.04983747397195082	
val_acc	0.9736	

Parameters (5)

Q Search parameters

Parameter	Value
model_architecture	MobileNetV2
epochs	5
batch_size	32
lr	0.001
dataset_size	17498

Logged models (1)

Model attributes

Type
Step
Model name
Status
Created
Registered models
Dataset
val_loss
train_acc

Output
0
model
Ready
6 minutes ago
-
-
0.07634218002557755
0.9585095439478796

3. FastAPI Deployment & Real-time Inference

A RESTful API was implemented using FastAPI to serve real-time predictions. The endpoint takes an uploaded image, resizes and normalizes the input tensor, and returns prediction probabilities.

- **Endpoint Tested:** POST /predict
- **Sample Input:** 3.jpg
- **Prediction Output:** Cat (Confidence: 99.996%)

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@3.jpg;type=image/jpeg'
```

Request URL

```
http://127.0.0.1:8000/predict
```

Server response

Code
Details

200

Response body

```
{
  "filename": "3.jpg",
  "prediction": "Cat",
  "confidence": 0.999666213989258,
  "probabilities": {
    "Cat": 0.999666213989258,
    "Dog": 0.00033424475986976177
  }
}
```

Download

Response headers

```
content-length: 143
content-type: application/json
date: Sat, 29 Aug 2026 08:50:48 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

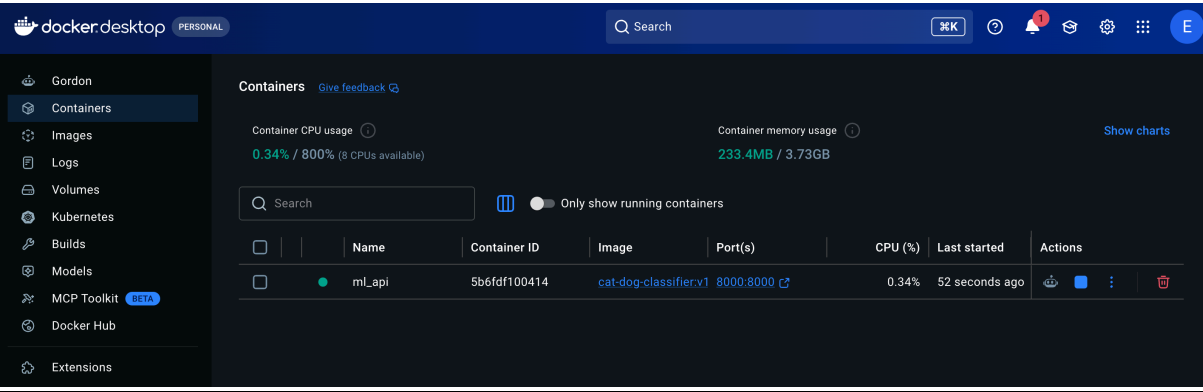
Controls Accept header.

Example Value
Schema

4. Containerization with Docker

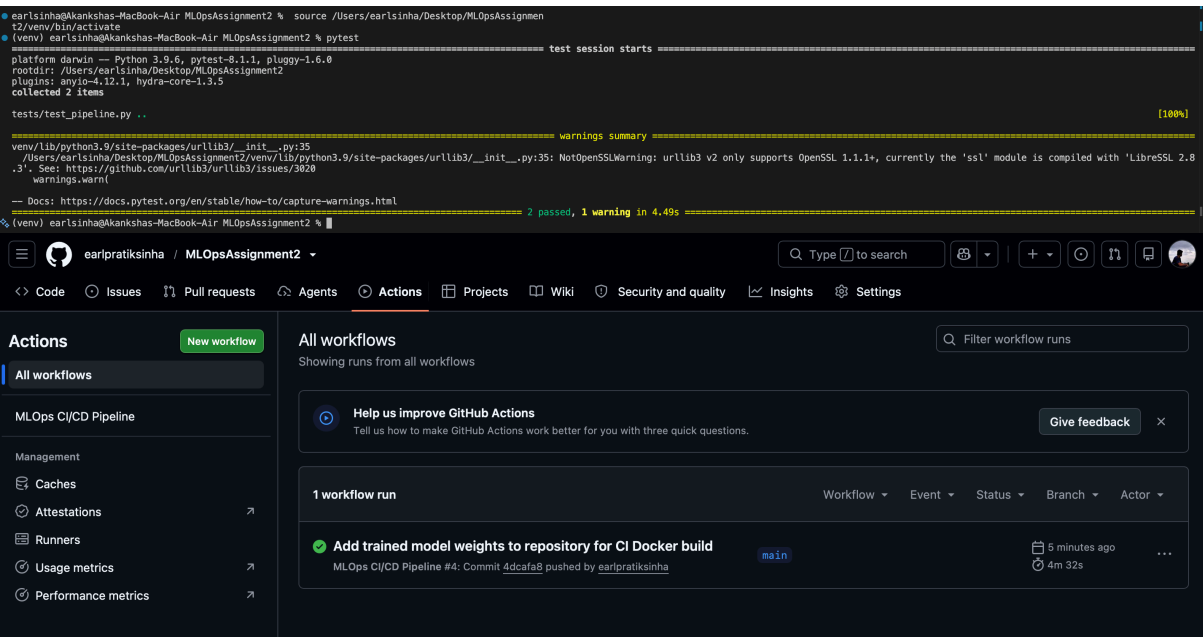
The application was containerized using a multi-stage Docker build with a Python 3.9 base image to ensure environment portability and isolation.

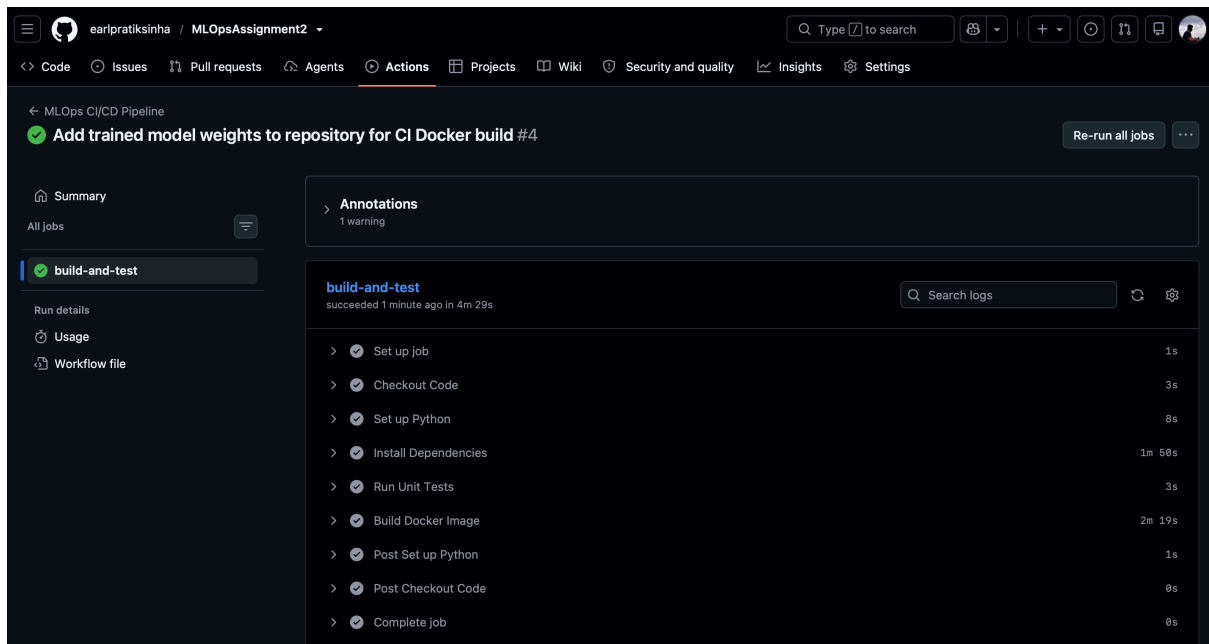
- **Container Image:** `cat-dog-classifier:v1`
- **Container Name:** `ml_api` (Container ID: `5b6fdf100414`)
- **Port Mapping:** Exposed locally on `0.0.0.0:8000->8000/tcp`



5. Automated Testing & Continuous Integration (CI) Unit tests were implemented using `pytest` to validate data preprocessing pipeline dimensions and image transformations. An automated CI workflow was configured via GitHub Actions to run tests and verify container builds on every push to the `main` branch.

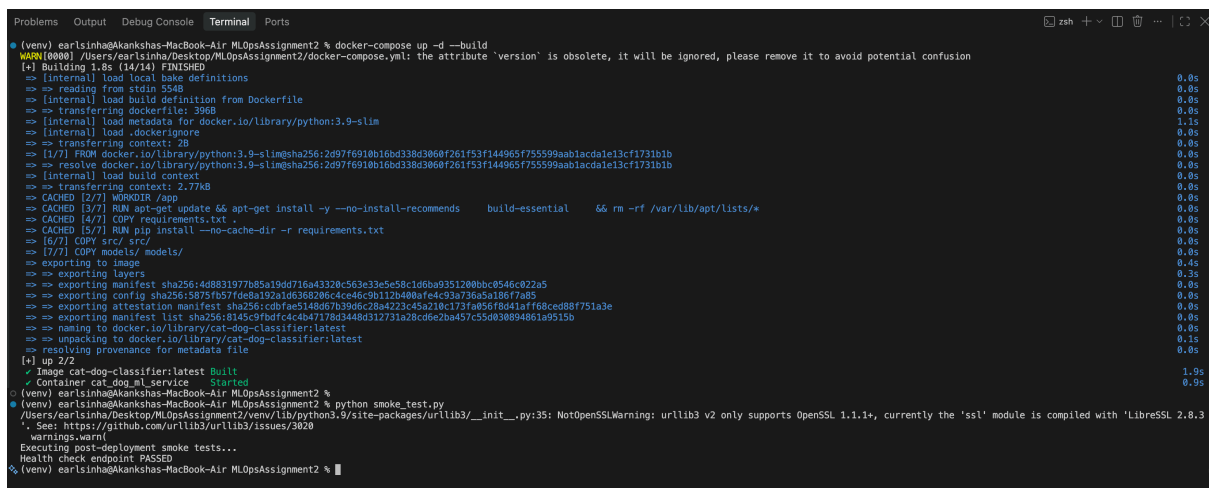
- **Unit Testing:** Executed `pytest tests/` (2 tests passed).
- **CI Workflow:** Configured `.github/workflows/ci.yml` for automated linting, testing, and Docker image builds.





6. Deployment, Health Smoke Testing & Monitoring (M4 & M5) Infrastructure deployment was configured using `docker-compose.yml` to orchestrate service delivery. The FastAPI application incorporates an explicit `/health` endpoint and structured request/latency logging. Post-deployment integrity is verified using an automated smoke testing script.

- **Deployment Manifest:** `docker-compose.yml` mapping exposed port `8000:8000`.
- **Smoke Testing:** Executed `python smoke_test.py` confirming `/health` endpoint status.
- **Logging & Monitoring:** Request metadata, prediction labels, and latency metrics logged to standard output.



Project Structure

MLOpsAssignment2/

```
├── src/
│   ├── app.py      # FastAPI service
│   └── train.py     # Model training
```

```
| └─ preprocess.py # Data preprocessing
|   └─ tests/
|     └─ test_pipeline.py
|       └─ models/
|         └─ best_model.pth
|           └─ data/
|             └─ raw/      # Original dataset (DVC tracked)
|               └─ processed/ # Preprocessed data
|                 └─ .github/workflows/
|                   └─ ci.yml
|                     └─ Dockerfile
|                       └─ docker-compose.yml
|                         └─ dvc.yaml
|                           └─ requirements.txt
```

Future Work

- Implement model monitoring with Prometheus/Grafana
 - Add data drift detection
 - Deploy to cloud (AWS/GCP) with autoscaling
 - Use ONNX for faster inference
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